

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / pm10
Observed / expected	72 / 28.4085
Time	2026-08-04 16:00 UTC
Location	29.6700, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	7 / 622	34.78 to 100; mean 73.2578	59.12 at 2026-08-04 15:55Z; dt 5 min
asos	temperature	degC	7 / 622	23.8889 to 36.6667; mean 29.5777	32 at 2026-08-04 15:55Z; dt 5 min
asos	wind_direction	degree	7 / 773	0 to 320; mean 143.389	260 at 2026-08-04 15:55Z; dt 5 min
asos	wind_speed	m/s	7 / 810	0 to 33.4389; mean 2.7723	3.08666 at 2026-08-04 15:55Z; dt 5 min
noaa_gfs	gh_500	m	8 / 80	5879.26 to 5926.86; mean 5904.75	5915.87 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	8 / 80	11.2803 to 2456.55; mean 697.483	2075.04 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	8 / 80	29.2 to 58.478; mean 37.4041	38.1486 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	8 / 80	1006.72 to 1015.12; mean 1010.69	1011.9 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	t_850	degC	8 / 80	18.2063 to 21.0207; mean 19.34	18.91 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	8 / 80	-3.62395 to 2.00478; mean -0.0099	0.106125 at 2026-08-04 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	8 / 80	0.378384 to 5.58564; mean 2.4917	2.36153 at 2026-08-04 18:00Z; dt 120 min
openaq	ozone	ppm	17 / 945	-0.004 to 0.078; mean 0.0225	0.026 at 2026-08-04 16:00Z; dt 0 min
openaq	pm10	ug/m3	3 / 182	13 to 172; mean 51.7582	72 at 2026-08-04 16:00Z; dt 0 min
openaq	pm25	ug/m3	72 / 2702	2.27 to 48.02; mean 11.542	23.8 at 2026-08-04 16:00Z; dt 0 min
openweather	cloud_cover	percent	5 / 324	0 to 100; mean 13.6296	0 at 2026-08-04 16:00Z; dt 0.3 min
openweather	humidity	percent	5 / 324	45 to 95; mean 73.892	70 at 2026-08-04 16:00Z; dt 0.3 min
openweather	precipitation	mm/h	5 / 324	0 to 0.39; mean 0.0065	0 at 2026-08-04 16:00Z; dt 0.3 min
openweather	pressure	hPa	5 / 324	1009 to 1017; mean 1012.78	1014 at 2026-08-04 16:00Z; dt 0.3 min
openweather	temperature	degC	5 / 324	25.09 to 37.35; mean 30.4719	32.13 at 2026-08-04 16:00Z; dt 0.3 min
openweather	wind_direction	degree	5 / 324	0 to 325; mean 180.676	151 at 2026-08-04 16:00Z; dt 0.3 min
openweather	wind_speed	m/s	5 / 324	0.45 to 6.47; mean 2.1614	2.24 at 2026-08-04 16:00Z; dt 0.3 min
purpleair	pm25	ug/m3	67 / 4164	0 to 557.7; mean 13.4716	16 at 2026-08-04 16:00Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_ch4_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_co_column	mol/m^2	5 / 5	0.0277646 to 0.0314233; mean 0.0303	0.0277646 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_co_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_hcho_column	mol/m^2	3 / 3	0.000121637 to 0.00030109; mean 0.0002	0.000121637 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_hcho_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_no2_column	mol/m^2	3 / 3	3.75274e-05 to 5.22023e-05; mean 0	3.75274e-05 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_no2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_o3_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_so2_column	mol/m^2	3 / 3	-3.3714e-05 to 0.000178122; mean 0.0001	-3.3714e-05 at 2026-08-04 20:10Z; dt 250.3 min
sentinel5p	s5p_so2_granule_available	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; dt 250.3 min
tceq	co	ppm	3 / 156	0.1 to 0.9; mean 0.2583	0.2 at 2026-08-04 16:00Z; dt 0 min
tceq	no2	ppb	12 / 633	-2.7 to 33.6; mean 5.0397	1.2 at 2026-08-04 16:00Z; dt 0 min
tceq	so2	ppb	3 / 154	-0.5 to 7.2; mean -0.0682	0 at 2026-08-04 16:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	humidity	08-03 00Z 79.85, 06Z 88.24, 12Z 68.41, 18Z 46.44, 08-04 00Z 75.34, 06Z 90.78, 12Z 70.42, 18Z 53.5, 08-05 00Z 79.38, 06Z 92.15, 12Z 70.64, 18Z 47.4
asos	temperature	08-03 00Z 28.09, 06Z 26.34, 12Z 30.64, 18Z 34.39, 08-04 00Z 28.59, 06Z 25.56, 12Z 30.45, 18Z 33.83, 08-05 00Z 28.82, 06Z 26.39, 12Z 30.36, 18Z 34.26
asos	wind_direction	08-03 00Z 180.8, 06Z 135.1, 12Z 162.6, 18Z 151.9, 08-04 00Z 157.8, 06Z 70.82, 12Z 136.5, 18Z 163.2, 08-05 00Z 166.4, 06Z 93.84, 12Z 168.4, 18Z 199.4
asos	wind_speed	08-03 00Z 2.387, 06Z 1.488, 12Z 2.275, 18Z 4.623, 08-04 00Z 3.374, 06Z 0.8376, 12Z 2.018, 18Z 5.535, 08-05 00Z 3.31, 06Z 1.232, 12Z 2.948, 18Z 3.887
noaa_gfs	gh_500	08-03 06Z 5887, 12Z 5880, 18Z 5896, 08-04 00Z 5900, 06Z 5914, 12Z 5900, 18Z 5916, 08-05 00Z 5918, 06Z 5926, 12Z 5911
noaa_gfs	pbl_height	08-03 06Z 167.6, 12Z 95.48, 18Z 2217, 08-04 00Z 938.1, 06Z 301.5, 12Z 101.9, 18Z 2014, 08-05 00Z 679.3, 06Z 323.4, 12Z 136.4
noaa_gfs	precipitable_water	08-03 06Z 31.42, 12Z 33.08, 18Z 34.9, 08-04 00Z 38.49, 06Z 33.44, 12Z 35.65, 18Z 39.53, 08-05 00Z 49.65, 06Z 38.7, 12Z 39.19
noaa_gfs	surface_pressure	08-03 06Z 1009, 12Z 1010, 18Z 1010, 08-04 00Z 1008, 06Z 1010, 12Z 1011, 18Z 1012, 08-05 00Z 1011, 06Z 1013, 12Z 1014
noaa_gfs	t_850	08-03 06Z 19.23, 12Z 18.47, 18Z 18.84, 08-04 00Z 19.88, 06Z 20, 12Z 19.02, 18Z 18.84, 08-05 00Z 19.2, 06Z 20.61, 12Z 19.3
noaa_gfs	u_10m	08-03 06Z 1.145, 12Z 1.115, 18Z -1.169, 08-04 00Z -2.435, 06Z 0.7441, 12Z 0.6063, 18Z -1.129, 08-05 00Z -0.2686, 06Z 1.071, 12Z 0.222

Source	Metric	Values
noaa_gfs	v_10m	08-03 06Z 2.214, 12Z 0.8743, 18Z 1.766, 08-04 00Z 4.523, 06Z 2.888, 12Z 1.277, 18Z 2.549, 08-05 00Z 4.38, 06Z 2.752, 12Z 1.695
openaq	ozone	08-03 00Z 0.04612, 06Z 0.02463, 12Z 0.02048, 18Z 0.05143, 08-04 00Z 0.02236, 06Z 0.009273, 12Z 0.01327, 18Z 0.02896, 08-05 00Z 0.01676, 06Z 0.009621, 12Z 0.0106, 18Z 0.02456
openaq	pm10	08-03 00Z 22, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 40.67, 12Z 73.56, 18Z 76.56, 08-05 00Z 56.13, 06Z 48.75, 12Z 72.87, 18Z 71.67
openaq	pm25	08-03 00Z 12.26, 06Z 9.387, 12Z 9.881, 18Z 12.12, 08-04 00Z 11.68, 06Z 15.27, 12Z 14.76, 18Z 11.79, 08-05 00Z 10.52, 06Z 11.23, 12Z 10.14, 18Z 8.753
openweather	cloud_cover	08-03 00Z 10.1, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13, 06Z 1.8, 12Z 0.3333, 18Z 23, 08-05 00Z 79.68, 06Z 9.8, 12Z 0.1333, 18Z 2
openweather	humidity	08-03 00Z 76.3, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7, 06Z 87.3, 12Z 77.63, 18Z 56.21, 08-05 00Z 75.52, 06Z 88.93, 12Z 78.93, 18Z 54.36
openweather	precipitation	08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0, 06Z 0, 12Z 0.01, 18Z 0, 08-05 00Z 0.02387, 06Z 0.03533, 12Z 0, 18Z 0
openweather	pressure	08-03 00Z 1010, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1013, 08-05 00Z 1013, 06Z 1015, 12Z 1017, 18Z 1016
openweather	temperature	08-03 00Z 29.36, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17, 06Z 26.51, 12Z 29.98, 18Z 35.38, 08-05 00Z 30.17, 06Z 27.29, 12Z 29.94, 18Z 35.47
openweather	wind_direction	08-03 00Z 195.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161, 06Z 175, 12Z 192.8, 18Z 163.8, 08-05 00Z 180.6, 06Z 194, 12Z 190.8, 18Z 179.1
openweather	wind_speed	08-03 00Z 2.316, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572, 06Z 1.775, 12Z 1.748, 18Z 3.054, 08-05 00Z 3.095, 06Z 1.718, 12Z 1.86, 18Z 2.426
purpleair	pm25	08-03 00Z 13.86, 06Z 11.94, 12Z 12.27, 18Z 13.73, 08-04 00Z 13.27, 06Z 17.57, 12Z 15.78, 18Z 13.68, 08-05 00Z 14.44, 06Z 13.24, 12Z 11.02, 18Z 8.618
sentinel5p	s5p_aer_ai_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_ch4_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_co_column	08-03 18Z 0.03095, 08-04 18Z 0.02776
sentinel5p	s5p_co_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_hcho_column	08-03 18Z 0.0002755, 08-04 18Z 0.0001216
sentinel5p	s5p_hcho_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_no2_column	08-03 18Z 4.973e-05, 08-04 18Z 3.753e-05
sentinel5p	s5p_no2_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_o3_granule_available	08-03 18Z 1, 08-04 18Z 1
sentinel5p	s5p_so2_column	08-03 18Z 0.0001056, 08-04 18Z -3.371e-05
sentinel5p	s5p_so2_granule_available	08-03 18Z 1, 08-04 18Z 1
tceq	co	08-03 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.2556, 12Z 0.3167, 18Z 0.2615, 08-05 00Z 0.3, 06Z 0.2222, 12Z 0.2722, 18Z 0.2
tceq	no2	08-03 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 4.78, 12Z 6.113, 18Z 3.278, 08-05 00Z 2.103, 06Z 3.751, 12Z 4.996, 18Z 2.381
tceq	so2	08-03 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2556, 12Z -0.16, 18Z -0.1077, 08-05 00Z -0.1, 06Z -0.2778, 12Z -0.2111, 18Z -0.2

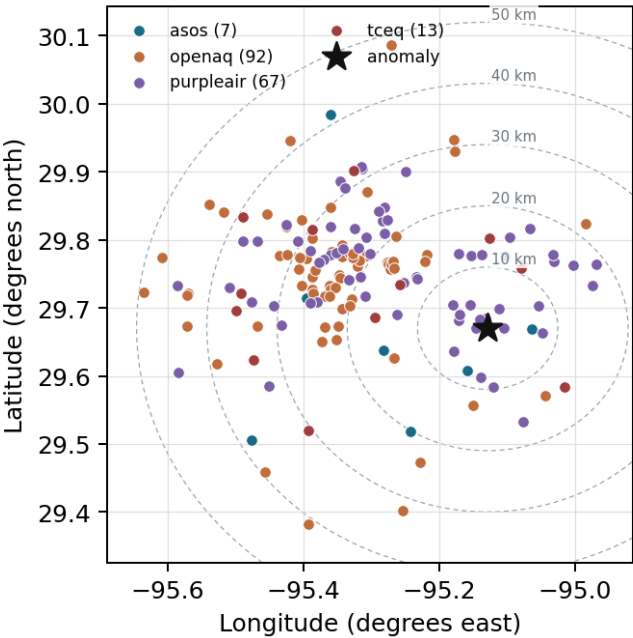
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

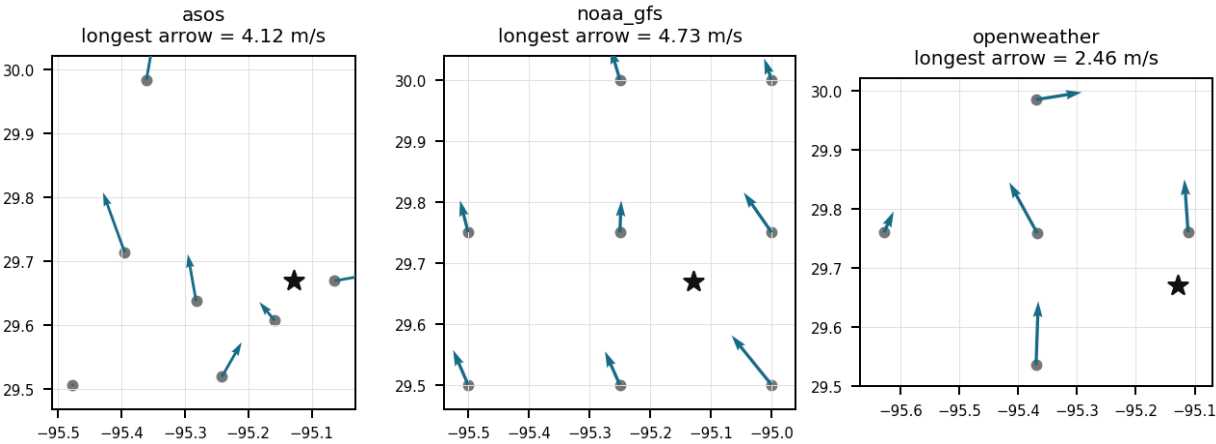
Source	Metric	Values
asos	humidity	T41 (6.224 km) 74.88, EFD (7.56 km) 73.96, HOU (15.304 km) 73.79, LVJ (20.054 km) 74.19, MCJ (26.202 km) 72.11, AXH (38.303 km) 72.74, IAH (41.517 km) 71.35
asos	temperature	T41 (6.224 km) 29.5, EFD (7.56 km) 29.78, HOU (15.304 km) 29.83, LVJ (20.054 km) 29.44, MCJ (26.202 km) 29.83, AXH (38.303 km) 29.04, IAH (41.517 km) 30.02
asos	wind_direction	T41 (6.224 km) 171.4, EFD (7.56 km) 136.5, HOU (15.304 km) 184.7, LVJ (20.054 km) 107.1, MCJ (26.202 km) 160.3, AXH (38.303 km) 70.56, IAH (41.517 km) 172.2
asos	wind_speed	T41 (6.224 km) 3.168, EFD (7.56 km) 3.853, HOU (15.304 km) 3.447, LVJ (20.054 km) 2.042, MCJ (26.202 km) 3.715, AXH (38.303 km) 0.9667, IAH (41.517 km) 2.819
noaa_gfs	gh_500	gfs:29.75,-95.25 (14.723 km) 5905, gfs:29.75,-95.00 (15.268 km) 5905, gfs:29.50,-95.25 (22.259 km) 5904, gfs:29.50,-95.00 (22.624 km) 5904, gfs:29.75,-95.50 (36.964 km) 5905, gfs:30.00,-95.25 (38.518 km) 5905, gfs:30.00,-95.00 (38.729 km) 5905, gfs:29.50,-95.50 (40.594 km) 5905
noaa_gfs	pbl_height	gfs:29.75,-95.25 (14.723 km) 862.2, gfs:29.75,-95.00 (15.268 km) 741.7, gfs:29.50,-95.25 (22.259 km) 606.3, gfs:29.50,-95.00 (22.624 km) 650.1, gfs:29.75,-95.50 (36.964 km) 833.1, gfs:30.00,-95.25 (38.518 km) 767.6, gfs:30.00,-95.00 (38.729 km) 474.4, gfs:29.50,-95.50 (40.594 km) 644.5
noaa_gfs	precipitable_water	gfs:29.75,-95.25 (14.723 km) 37.6, gfs:29.75,-95.00 (15.268 km) 36.78, gfs:29.50,-95.25 (22.259 km) 38.49, gfs:29.50,-95.00 (22.624 km) 37.1, gfs:29.75,-95.50 (36.964 km) 37.44, gfs:30.00,-95.25 (38.518 km) 37.39, gfs:30.00,-95.00 (38.729 km) 36.18, gfs:29.50,-95.50 (40.594 km) 38.24
noaa_gfs	surface_pressure	gfs:29.75,-95.25 (14.723 km) 1011, gfs:29.75,-95.00 (15.268 km) 1012, gfs:29.50,-95.25 (22.259 km) 1011, gfs:29.50,-95.00 (22.624 km) 1012, gfs:29.75,-95.50 (36.964 km) 1010, gfs:30.00,-95.25 (38.518 km) 1009, gfs:30.00,-95.00 (38.729 km) 1010, gfs:29.50,-95.50 (40.594 km) 1010
noaa_gfs	t_850	gfs:29.75,-95.25 (14.723 km) 19.29, gfs:29.75,-95.00 (15.268 km) 19.25, gfs:29.50,-95.25 (22.259 km) 19.34, gfs:29.50,-95.00 (22.624 km) 19.31, gfs:29.75,-95.50 (36.964 km) 19.49, gfs:30.00,-95.25 (38.518 km) 19.34, gfs:30.00,-95.00 (38.729 km) 19.21, gfs:29.50,-95.50 (40.594 km) 19.49
noaa_gfs	u_10m	gfs:29.75,-95.25 (14.723 km) 0.3147, gfs:29.75,-95.00 (15.268 km) -0.1523, gfs:29.50,-95.25 (22.259 km) -0.02427, gfs:29.50,-95.00 (22.624 km) 0.01673, gfs:29.75,-95.50 (36.964 km) 0.1557, gfs:30.00,-95.25 (38.518 km) -0.3533, gfs:30.00,-95.00 (38.729 km) 0.1207, gfs:29.50,-95.50 (40.594 km) -0.1573
noaa_gfs	v_10m	gfs:29.75,-95.25 (14.723 km) 2.439, gfs:29.75,-95.00 (15.268 km) 2.434, gfs:29.50,-95.25 (22.259 km) 2.547, gfs:29.50,-95.00 (22.624 km) 2.933, gfs:29.75,-95.50 (36.964 km) 2.758, gfs:30.00,-95.25 (38.518 km) 2.078, gfs:30.00,-95.00 (38.729 km) 2.248, gfs:29.50,-95.50 (40.594 km) 2.498
openaq	ozone	332 (0.0 km) 0.02433, 2800 (11.568 km) 0.02257, 2039 (14.072 km) 0.02162, 320 (14.284 km) 0.01722, 3378 (14.341 km) 0.02224, 5077791 (14.578 km) 0.01758, 3214 (14.733 km) 0.024, 325 (16.161 km) 0.006146, +9 more
openaq	pm10	271 (0.0 km) 45.83, 13380082 (14.341 km) 75.25, 15852419 (21.343 km) 34.29
openaq	pm25	268 (0.0 km) 17.22, 13955815 (12.759 km) 9.707, 14404954 (13.75 km) 11.56, 14157975 (14.287 km) 14.21, 3379 (14.341 km) 16.23, 15539682 (14.345 km) 12.09, 8967123 (14.399 km) 13.8, 8967479 (14.399 km) 12.48, +64 more
openweather	cloud_cover	grid:east (10.189 km) 10.26, grid:center (25.125 km) 13.78, grid:south (27.694 km) 14.48, grid:north (42.053 km) 14.4, grid:west (49.32 km) 15.23
openweather	humidity	grid:east (10.189 km) 77.4, grid:center (25.125 km) 72.56, grid:south (27.694 km) 74.15, grid:north (42.053 km) 71.88, grid:west (49.32 km) 73.45
openweather	precipitation	grid:east (10.189 km) 0.003692, grid:center (25.125 km) 0.01109, grid:south (27.694 km) 0.006, grid:north (42.053 km) 0.01169, grid:west (49.32 km) 0
openweather	pressure	grid:east (10.189 km) 1013, grid:center (25.125 km) 1013, grid:south (27.694 km) 1013, grid:north (42.053 km) 1013, grid:west (49.32 km) 1013
openweather	temperature	grid:east (10.189 km) 30.21, grid:center (25.125 km) 31.01, grid:south (27.694 km) 30.17, grid:north (42.053 km) 30.51, grid:west (49.32 km) 30.48
openweather	wind_direction	grid:east (10.189 km) 182.6, grid:center (25.125 km) 198.1, grid:south (27.694 km) 185.5, grid:north (42.053 km) 152.2, grid:west (49.32 km) 185.3
openweather	wind_speed	grid:east (10.189 km) 3.226, grid:center (25.125 km) 2.143, grid:south (27.694 km) 1.925, grid:north (42.053 km) 2.003, grid:west (49.32 km) 1.51
purpleair	pm25	128007 (1.624 km) 6.748, 161689 (1.695 km) 13.67, 222589 (1.835 km) 14.42, 268165 (2.295 km) 15.43, 222593 (3.625 km) 13.63, 186263 (4.256 km) 15.63, 98633 (4.617 km) 6.592, 244939 (4.67 km) 10.97, +59 more
tceq	co	482011039 (0.032 km) 0.2037, 482011035 (14.34 km) 0.1638, 482011052 (29.74 km) 0.3927

Source	Metric	Values
tceq	no2	482011039 (0.032 km) 2.685, 482011015 (10.969 km) 5.864, 482011035 (14.34 km) 9.982, 482011050 (14.586 km) 1.198, 482010026 (14.755 km) 7.555, 482010416 (16.16 km) 4.535, 482011052 (29.74 km) 15.79, 480391004 (30.466 km) 2.44, +4 more
tceq	so2	482011039 (0.032 km) 0.3111, 482011035 (14.34 km) -0.3545, 482010051 (33.811 km) -0.2089

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The temperature on August 4th, 2026 was relatively high, with a mean value of 30.47°C, which may have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 2 of 36

There is no evidence to suggest that any other factors such as wind direction or speed, precipitation, or atmospheric pressure played a significant role in the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 3 of 36

Industrial bulk material handling, earthmoving, or construction operations in the Deer Park and Houston Ship Channel industrial sector generated localized coarse particulate emissions observed by openaq and tceq.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 4 of 36

The openaq PM10 measurement at coordinates 29.670, -95.129 was 72.0 ug/m3 at 2026-08-04T16:00:00+00:00, compared with an expected value of 28.41 ug/m3 and a z-score of 4.36, and the anomaly was labeled severe.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 5 of 36

Longer-range transported dust is a secondary hypothesis because the synoptic pattern was hot and mostly cloud-free, but the available satellite trace-gas columns did not show a corresponding combustion signature and the evidence provided does not specifically demonstrate a regional dust plume over Houston at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 6 of 36

OpenAQ measurements at the anomaly site recorded a PM10 concentration of 72.0 ug/m3 alongside a concurrent PM2.5 level of 23.8 ug/m3, demonstrating that coarse fraction particles comprised most of the particulate mass.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 7 of 36

The air quality anomaly is related to PM2.5, which is elevated.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 8 of 36

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:

☐ V

☐ I

☐ U

Note:

Claim 9 of 36

The anomaly occurred on August 4th, around 16:00 UTC.

Mark exactly one:

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Note:

Claim 10 of 36

Local mechanically generated dust or resuspended road or construction material is supported because PM10 at the anomaly site reached 72.0 ug/m3 while co-located PM2.5 was only 23.8 ug/m3 and nearby PurpleAir PM2.5 was about 16.0 ug/m3, a pattern more consistent with coarse particles than with a fine-particle combustion plume.

Mark exactly one:

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Note:

Claim 11 of 36

The Houston anomaly occurred under hot, sunny daytime conditions favorable for dust resuspension and vertical mixing because asos and openweather showed temperatures near 32 C, humidity near 59 to 70 percent, cloud cover near 0 percent, and gfs indicated afternoon planetary boundary layer height near 2075 m.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 12 of 36

Regional advection of coarse crustal aerosols across coastal Texas contributed to elevated baseline coarse particulate matter concentrations measured by openaq and purpleair.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 13 of 36

Resuspended local dust is a leading hypothesis because PM10 rose strongly while nearby PM2.5 and co-located combustion tracers remained much lower, a pattern that is more consistent with coarse particles from roads, bare soil, or disturbed surfaces than with smoke or fresh exhaust.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 14 of 36

The PM10 anomaly recorded in Houston at 16:00 UTC on August 4, 2026, was dominated by coarse fraction particulate matter while PM2.5 levels remained comparatively low.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 15 of 36

An elevated PM10 concentration of 72.0 ug/m3 was recorded on August 4, 2026, at 16:00 UTC at coordinates (29.670, -95.129), representing a severe anomaly with a z-score of 4.36 relative to an expected baseline of 28.41 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 16 of 36

The temperature was unusually high on August 4th, leading to increased emissions from industrial sources and vehicles.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 17 of 36

Localized industrial, port, construction, or unpaved-road material handling is a plausible cause because the PM10 enhancement was strong in a sparse PM10 network, the anomaly site had only light-to-moderate winds during a hot daytime period, and those conditions can produce short-lived coarse-particle plumes near sources.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 18 of 36

The PM2.5 levels were higher than usual in the surrounding area, potentially due to increased particulate matter emissions from local sources or distant wildfires.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 19 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 20 of 36

The anomaly time and location were 2026-08-04T16:00:00+00:00 at coordinates 29.670, -95.129 in the Houston, Texas area.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 21 of 36

Local mechanically generated dust or resuspended road or construction material is further supported because the anomaly occurred during hot daytime conditions with low cloud cover and only modest winds, with about 32 C temperature, about 59 percent relative humidity at the nearest ASOS station, 0 percent cloud cover near the event, and wind speeds near 2 to 3 m/s, which are conditions that can dry surfaces and allow localized dust emissions to persist.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 22 of 36

OpenAQ PM10 monitoring in the area recorded 6-hour mean concentrations peaking at 73.56 ug/m3 at 12:00 UTC and 76.56 ug/m3 at 18:00 UTC on August 4, 2026.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 23 of 36

TCEQ monitoring at the nearest station recorded minimal co-pollutant concentrations of 1.2 ppb NO2 and 0.0 ppb SO2 at the time of the event, indicating that localized industrial combustion plumes were unlikely primary drivers.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 24 of 36

Localized wind-blown dust resuspension from unpaved surfaces and dry surface soils driven by daytime heating and surface winds contributed to the PM10 spike recorded by openaq.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 25 of 36

The location of the anomaly is near sensor 128007 (1.624 km) and monitor 482011039 (0.032 km).

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 26 of 36

A strong wind pattern on August 5th may have dispersed pollutants from nearby refineries and industrial areas towards the anomaly location.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 27 of 36

Regional transported smoke or a chemically rich industrial combustion plume is contradicted because ozone, CO, NO₂, and SO₂ were not strongly elevated near the anomaly time and same-day satellite columns for CO, NO₂, HCHO, and SO₂ were not enhanced, although transported mineral dust remains a weaker possibility because PM₁₀ stayed elevated across multiple 6-hour periods and southerly to southeasterly low-level flow was present.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 28 of 36

Low concurrent concentrations of nitrogen dioxide, sulfur dioxide, and carbon monoxide rule out major industrial combustion or fire plumes as the source of the PM₁₀ spike.

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Note:

Claim 29 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM_{2.5}.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 30 of 36

At the time of the PM10 anomaly on August 4, 2026, at 16:00 UTC, the nearest PM2.5 sensor recorded a concentration of 23.8 ug/m3, demonstrating that coarse particulate matter composed the majority of the elevated PM10 mass.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 31 of 36

The Houston anomaly is unlikely to be primarily a combustion plume because nearby gaseous copollutants were not elevated at the anomaly time, with tceq showing CO at 0.2 ppm, NO2 at 1.2 ppb, and SO2 at 0.0 ppb near the site.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 32 of 36

At 2026-08-04T16:00:00+00:00 at the anomaly location, openaq PM2.5 was 23.8 ug/m3, openaq ozone was 0.026 ppm, tceq CO was 0.2 ppm, tceq NO2 was 1.2 ppb, and tceq SO2 was 0.0 ppb.

Mark exactly one:	☐ V	☐ I	☐ U
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Note:

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:

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Note:

Claim 34 of 36

Daytime planetary boundary layer height exceeding 2000 meters accompanied by warm temperatures and low cloud cover facilitated localized surface dust resuspension.

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Note:

Claim 35 of 36

OpenAQ network data showed broad PM10 elevation across the region with 6-hour mean values reaching 73.56 ug/m3 to 76.56 ug/m3, supporting widespread coarse aerosol advection or dust resuspension rather than a single isolated point source.

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Note:

Claim 36 of 36

The Houston anomaly is most consistent with a local mechanically generated coarse-particle source because openaq recorded PM10 at 72 ug/m3 against an expected 28.4 ug/m3 at 2026-08-04T16:00Z while co-located PM2.5 was only 23.8 ug/m3.

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Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.
